

Mohammed Yusuf

System Engineer — Java Backend

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Summary

Backend Software Engineer with 3+ years of experience designing and delivering high-availability, distributed Java and Spring Boot systems at enterprise scale. Proven track record in event-driven microservices architecture with Apache Kafka, test-driven development with SonarQube quality gates, and CI/CD pipeline automation. Hands-on with relational databases, observability tooling, and production infrastructure. Microsoft Azure Security Engineer certified.

Technical Skills

Languages	Java, Python, SQL
Backend	Spring Boot, Spring Framework, Spring Data, Spring Security, Hibernate / JPA, Apache Kafka, REST API, Microservices, Event-Driven Architecture
Testing & Quality	JUnit, Mockito, TDD, Integration Testing, SonarQube, Code Review
DevOps & CI/CD	Docker, Jenkins, GitHub Actions, Maven, Linux, Oracle Cloud (OCI)
Databases	PostgreSQL, MySQL, SQLite, NoSQL (familiarity)
Observability	Prometheus, Grafana, Alertmanager, Distributed Tracing, Structured Logging

Experience

Tata Consultancy Services

Bengaluru, India

System Engineer — Java Backend

Aug 2022 – Present

- Designed and developed Java and Spring Boot microservices for mission-critical financial transaction platforms, engineering 40+ REST API endpoints and improving application execution speed by **35%**.
- Architected an event-driven, high-throughput pipeline using Apache Kafka, replacing synchronous REST calls and reducing P99 processing latency by **35%** while improving system throughput by **40%** — handling peak financial transaction loads with high availability.
- Implemented TDD using JUnit and Mockito with high unit test coverage and SonarQube quality gates enforced on every release — reducing production defects by **30%** across consecutive releases.
- Built and maintained CI/CD pipelines with Docker and Jenkins, reducing build and deployment cycle times by **20%** and enabling continuous delivery across test, integration, and production environments.
- Led migration of legacy .NET and mainframe systems to a scalable, distributed Spring Boot microservices architecture, delivering zero data loss during cutover across multiple production services.
- Conducted regular code reviews, drove continuous refactoring, and mentored junior engineers on Java best practices, accelerating team delivery velocity by 20%.
- Designed and documented RESTful API specifications, system architecture diagrams, and technical design documents for cross-functional teams, ensuring alignment between business requirements and scalable software solutions.
- Collaborated with product and QA teams in Agile/Scrum sprints — participating in story refinement, sprint planning, and retrospectives — consistently delivering features on time with zero critical post-release incidents.
- Optimised database query performance using Hibernate/JPA with PostgreSQL, reducing average query execution time through indexing strategies, lazy loading, and connection pool tuning across high-traffic service endpoints.

Projects

Falak — Production-Grade Self-Hosted Infrastructure

2025 – Present

Docker, Traefik v3, Prometheus, Grafana, Alertmanager, CrowdSec, Authelia, Ubuntu Server

- Designed and operate a high-availability 3-node cluster running 22 production services on Ubuntu with Docker, Traefik v3 reverse proxy, automated TLS, and full observability via Prometheus, Grafana, and Alertmanager with custom alert rules.
- Implemented structured logging, distributed monitoring, and automated alerting — achieving RPO 15min / RTO 2min via automated failover and AES-256 encrypted offsite backups.
- Hardened platform security with CrowdSec threat detection, Authelia SSO with TOTP 2FA, and a default-deny firewall — applying defence-in-depth across all services.
- Built workflow automation using n8n for scheduled tasks, Telegram-based alerting, and service health notifications — reducing manual operational overhead across the infrastructure.

Education & Certifications

AVS Engineering College — B.E. Mechanical Engineering, CGPA 7.49

Tamil Nadu, 2016–2020

Microsoft Certified: Azure Security Engineer Associate

2023–2027

Microsoft | Credential ID: C2BED8FD1B1373F